

ABSTRACT - Weather Forecast Viewer

The Weather Forecast Viewer is a Python-based web application developed using the Django framework to provide users with real-time and forecasted weather information through an interactive and responsive interface. The system integrates external weather data services, such as the OpenWeatherMap API, to fetch accurate meteorological data including temperature, humidity, wind speed, and atmospheric conditions for various locations.

Django is utilized to manage the backend logic, URL routing, and template rendering, enabling a structured and scalable web architecture. The application allows users to search for weather details of different cities and view short-term forecasts in an organized format. Data visualization is enhanced using libraries such as Matplotlib or Plotly to present trends and comparisons effectively.